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EVERYTHING I KNOW ABOUT TIME SERIES

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Difference Equations

The whole point of time series analysis is to uncover and describe the statistical properties and structure of a process across time. Certain processes may have a lot of structure, so much that advanced non-linear methods such as those enabled by machine learning are needed. Others have very little, and can be adequately described by a very simple model. As we are always worried about overfitting, and there is a general philosophy that parsimony is a good thing, we begin by studying the simplest of processes.

Assume we have the deterministic process

$$y_t = \phi y_{t-1} + w_t \quad (1)$$

In order to find the state of this process at any time t , we need some kind of initial condition. We may take y_{-1} as given, or if we want to look some arbitrary time setp j into the future from a given time t , we may take y_{t-1} as given. In either case, we can use recursion to find the statme of y_t at any future time t :

$$\begin{aligned} y_t &= \phi y_{t-1} + w_t, \\ y_{t+1} &= \phi y_t + w_{t+1} &= \phi(\phi y_{t-1} + w_t) + w_{t+1} &= \phi^2 y_{t-1} + \phi w_t + w_{t+1}, \\ y_{t+2} &= \phi y_{t+1} + w_{t+2} &= \phi(\phi^2 y_{t-1} + \phi w_t + w_{t+1}) + w_{t+2} &= \phi^3 y_{t-1} + \phi^2 w_t + \phi w_{t+1} + w_{t+2}, \\ &\vdots \\ y_{t+j} &= \phi^{j+1} y_{t-1} + \sum_{k=0}^j \phi^{j-k} w_{t+k}. \end{aligned}$$

The last line shows the general form of this process at some future state: it has the classic characteristic of a time series process in that its state at any point in time depends on two things: an initial condition y_{t-1} that encapsulates all of its history up until that point, and the shocks w_{t+k} that have affected the process since the initial condition. The effect of a shock at time t on the system at some future time $t + j$ is

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^j, \quad (2)$$

and the effect of the initial condition is

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^{j+1}. \quad (3)$$

So, it is clear that the stability of the process depends on ϕ . If $-1 < \phi < 1$, the effect of the initial condition and all shocks at some future horizon j goes to zero; if $1 < \phi < -1$, the effect of each compounds over time and the process explodes; if $\phi = \pm 1$, the shocks and the initial condition have a permanent effect of the process.

Now, instead of having y_t depend on only its previous value and a contemporary shock, we'll have it depend on the previous p values

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + w_t.$$

It is often more convenient to transform this into a first-order vector-valued equation,

$$\xi_t = F \xi_{t-1} + v_t$$

where

$$\xi_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix}$$

is $p \times 1$,

$$F = \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

is $p \times p$, and

$$v_t = \begin{pmatrix} w_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

By a similar recursive operation before, if we have an initial condition at ξ_{t-1} we can solve for the state of the process at some future horizon j ,

$$\begin{aligned}
\boldsymbol{\zeta}_t &= F\boldsymbol{\zeta}_{t-1} + \boldsymbol{v}_t, \\
\boldsymbol{\zeta}_{t+1} &= F\boldsymbol{\zeta}_t + \boldsymbol{v}_t &= F(F\boldsymbol{\zeta}_{t-1} + \boldsymbol{v}_t) + \boldsymbol{v}_{t+1} &= F^2\boldsymbol{\zeta}_{t-1} + F\boldsymbol{v}_t + \boldsymbol{v}_{t+1}, \\
\boldsymbol{\zeta}_{t+2} &= F\boldsymbol{\zeta}_{t+1} + \boldsymbol{v}_{t+2} &= F(F^2\boldsymbol{\zeta}_{t-1} + F\boldsymbol{v}_t + \boldsymbol{v}_{t+1}) + \boldsymbol{v}_{t+2} &= F^3\boldsymbol{\zeta}_{t-1} + F^2\boldsymbol{v}_t + F\boldsymbol{v}_{t+1} + \boldsymbol{v}_{t+2}, \\
&\vdots \\
\boldsymbol{\zeta}_{t+j} &= F^{j+1}\boldsymbol{\zeta}_{t-1} + \sum_{k=0}^j F^{j-k}\boldsymbol{v}_{t-k}
\end{aligned}$$

Note that the future value of interest y_{t+j} is the first row of this last equation. If f_{ij}^n is the (i, j) element of F^n , then

$$y_{t+j} = f_{11}^{j+1}y_{t-1} + f_{12}^{j+1}y_{t-2} + \cdots + f_{1p}^{j+1}y_{t-p} + w_{t+j} + f_{11}w_{t+j-1} + \cdots + f_{11}^jw_t$$